Gradients of Straight Lines

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 79 minutes · Total marks: 79

KEY RULES

Gradient	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Equation	$y = mx + c$
Parallel lines	Equal gradients: $m_1 = m_2$
Perpendicular	$m_1 \times m_2 = -1$
Through a point	$y - y_1 = m(x - x_1)$

WORKED EXAMPLE 1

[2 marks]

Find the gradient of the line through $(1, 3)$ **and** $(5, 11)$**.**

Step 1 Gradient = change in y ÷ change in x $\frac{11 - 3}{5 - 1} = \frac{8}{4}$

Answer: 2

WORKED EXAMPLE 2

[3 marks]

Find the equation of the line parallel to $y = 3x - 2$ **that passes through** $(2, 7)$**.**

Step 1 Parallel lines have the same gradient $y = 3x + c$

Step 2 Substitute the point $7 = 3 \times 2 + c$

Step 3 Solve for c $c = 1$

Answer: $y = 3x + 1$

WORKED EXAMPLE 3

[2 marks]

Write down the gradient of a line perpendicular to $y = 2x + 1$**.**

Step 1 Perpendicular gradients multiply to -1 $2 \times m = -1$

Answer: $-\frac{1}{2}$

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1 Find the gradient of the straight line with equation $5x + 2y = 7$

.....

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2 Point A has coordinates $(-3, 4)$
Point B has coordinates $(2, -6)$

Work out the gradient of the line AB

.....

(Total for question = 2 marks)

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3 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

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4 Line L has equation $y = 2x + 3$.

Find the equation of the line parallel to L that passes through $(1, -4)$.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

5 Find the equation of the line perpendicular to $y = \frac{1}{2}x + 1$ that passes through $(2, 5$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

6 Find the coordinates of the point where the lines $y = 2x + 1$ and $y = -x + 7$ meet.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

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Gradients of Straight Lines

7 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

.....

(2)

(b) Find the coordinates of the point where **L** crosses the y-axis.

(..... , ...)

(1)

(Total for question = 3 marks)

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8 The straight line $\mathbf{L}_1$ has equation $x + 2y = 4$
The straight line $\mathbf{L}_2$ passes through the points $(-1, -7)$ and $(7, 9)$

Michael says that the lines $\mathbf{L}_1$ and $\mathbf{L}_2$ are perpendicular.

Is Michael correct?
You must show clearly how you get your answer.

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9 The straight line **L** has equation $4y - 10x - 5 = 0$

(a) Find the gradient of **L**

.....

(2)

(b) Find the coordinates of the point where **L** crosses the y-axis.

(..... , ...)

(1)

(Total for question = 3 marks)

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10 Point A has coordinates (5, 8)
Point B has coordinates (9, –4)

(a) Work out the gradient of AB.

.....

(2)

The straight line **L** has equation $y = -4x + 5$

(b) Write down the gradient of a straight line that is perpendicular to **L**.

.....

(1)

(Total for question = 3 marks)

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11 A is the point with coordinates $(-5, 12)$

B is the point with coordinates $(19, -48)$

Find an equation of the straight line that passes through the points A and B

.....

(Total for question = 3 marks)

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12 A is the point with coordinates $(1, 3)$
B is the point with coordinates $(-2, -1)$

The line **L** has equation $3y = 4 - 2x$

Is line **L** parallel to AB?
Show your working clearly.

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13 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form $y = mx + c$

.....

(Total for question = 3 marks)

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14 The straight line $\mathbf{L}_1$ has equation $y = 6 - 2x$
The straight line $\mathbf{L}_2$ is perpendicular to $\mathbf{L}_1$ and passes through the point (4, 7)

Find the coordinates of the point where the line $\mathbf{L}_2$ crosses the x-axis.

(..... , ...)

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15 The straight line $\mathbf{L}_1$ has equation $2y = 6x - 5$

The straight line $\mathbf{L}_2$ is perpendicular to $\mathbf{L}_1$ and passes through the point $(9, -1)$

Find an equation for $\mathbf{L}_2$
Give your answer in the form $ay + bx = c$

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2019 · Question 15

16 **17** The straight line **L** passes through the points $(4, -1)$ and $(6, 4)$

The straight line **M** is perpendicular to **L** and intersects the y-axis at the point $(0, 8)$

Find the coordinates of the point where **M** intersects the x-axis.

(..... , ...)

(Total for Question 17 is 4 marks)

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17 The straight line **L** is perpendicular to the straight line with equation $2x + y = 9$ and passes through the point with coordinates $(8, 11)$

Find an equation for **L**
Give your answer in the form $y = mx + c$

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CHALLENGE QUESTIONS

18 Line **L** has equation $4y - 6x = 33$
Line **M** goes through the point $A\ (5, 6)$ and the point $B\ (-4, k)$

L is perpendicular to **M**.

Work out the value of k.

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19 (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

.....

(1)

The line **L** passes through the points with coordinates (–3, 1) and (2, –2)

(b) Find an equation of the line that is perpendicular to **L** and passes through the point with coordinates (–6, 4)
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

.....

(4)

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20 $PQRS$ is a square.

PR is a diagonal of the square.

P is the point with coordinates (4, 7)
R is the point with coordinates (8, −5)

Find an equation of the straight line that passes through the points Q and S
Give your answer in the form $ay = bx + c$ where a, b and c are integers.

.....

(Total for question = 5 marks)

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21 P and Q are two points.

The coordinates of P are (–1, 6)

The coordinates of Q are (5, –4)

Find an equation of the perpendicular bisector of PQ.
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

.....

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22 The straight line **L** passes through point A $(-6, 2)$ and point B $(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B.
The line **M** intersects the line $x = -1$ at point C.

Calculate the area of triangle ABC.

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TOTAL FOR PAPER IS 79 MARKS

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